

TORPSHAMMAR, Sweden

WMO: 023430

Lat: 62.494N

Lon: 16.277E

Elev: 100

StdP: 100.13

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.7	(c) -22.6	(d) -28.4	(e) 0.3	(f) -25.6	(g) -25.0	(h) 0.4	(i) -22.5	(j) 5.5	(k) 2.0	(l) 4.7	(m) 1.1	(n) 0.1	(o) 280	(p) 0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	26.9	17.9	24.8	16.9	22.8	16.0	19.3	24.3	18.1	22.9	17.0	21.2	1.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.6	21.1	16.2	11.7	19.9	15.2	10.9	18.9	55.2	24.3	51.5	23.0	47.9	21.3	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.5	(b) 4.6	(c) 3.9	(e) -28.9	(f) 29.2	(g) 3.0	(h) 1.9	(i) -31.0	(j) 30.5	(k) -32.7	(l) 31.6	(m) -34.4	(n) 32.6	(o) -36.5	(p) 34.0		
(a) 5.5	(b) 4.6	(c) 3.9	(e) -28.9	(f) 21.0	(g) 2.7	(h) 1.4	(i) -30.6	(j) 22.0	(k) -32.2	(l) 22.8	(m) -33.8	(n) 23.6	(o) -35.8	(p) 24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-8.2	-6.8	-2.0	3.4	8.8	12.9	15.9	14.4	9.6	3.5	-1.7	-6.1		
	DBStd	9.44	6.82	6.48	4.88	3.11	3.61	3.07	2.94	3.05	3.30	3.64	4.59	6.75		
	HDD10.0	2781	563	471	372	198	68	7	1	3	47	202	351	499		
	HDD18.3	5358	821	704	630	448	297	163	88	127	263	459	601	757		
	CDD10.0	483	0	0	0	1	29	96	183	139	34	2	0	0		
	CDD18.3	18	0	0	0	0	0	2	12	4	0	0	0	0		
	CDH23.3	340	0	0	0	0	20	74	184	62	0	0	0	0		
	CDH26.7	62	0	0	0	0	1	11	44	7	0	0	0	0		
(14) Wind	WSAvg	1.3	0.9	1.1	1.5	1.6	1.7	1.9	1.5	1.2	1.2	1.2	1.0	0.9		
(15) Precipitation	PrecAvg	644	42	28	30	37	44	66	92	88	56	61	52	48		
	PrecMax	911	86	57	67	104	108	114	229	167	168	157	130	78		
	PrecMin	491	6	3	6	3	11	27	15	46	14	18	17	5		
	PrecStd	101	22	13	14	25	24	25	49	38	35	33	29	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	7.6	11.6	19.6	25.8	28.2	30.4	27.7	21.6	15.6	10.2	6.9		
		MCWB	4.0	4.0	6.5	11.1	15.2	17.1	19.4	18.8	15.5	11.9	8.7	4.8		
	2%	DB	3.6	5.8	8.8	15.3	22.6	25.6	27.8	25.1	19.4	12.7	7.7	4.9		
		MCWB	2.0	3.3	4.6	8.6	13.6	16.1	18.7	18.2	14.5	10.0	6.5	3.4		
	5%	DB	1.8	3.9	6.9	12.9	19.8	23.0	25.3	22.9	17.7	11.1	5.7	3.3		
		MCWB	0.8	1.9	3.4	7.1	12.2	14.9	17.8	17.1	13.6	9.2	4.8	2.2		
	10%	DB	0.4	1.6	5.1	10.7	17.1	20.6	23.1	20.7	15.7	9.6	4.3	1.7		
		MCWB	-0.2	0.3	2.3	6.0	11.0	13.9	16.7	16.1	12.5	8.0	3.4	0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.4	6.9	11.8	16.1	18.8	21.1	20.5	16.5	12.6	9.1	5.1		
		MCDB	5.7	6.7	11.3	19.4	23.6	25.5	26.9	24.7	20.5	14.8	10.1	6.2		
	2%	WB	2.3	3.4	5.1	9.2	14.5	16.7	19.7	19.1	15.1	10.7	6.6	3.6		
		MCDB	3.3	5.6	8.4	14.3	21.1	22.5	25.2	23.1	18.4	12.0	7.6	4.8		
	5%	WB	0.8	2.1	3.7	7.7	12.9	15.6	18.6	17.8	14.1	9.5	5.0	2.2		
		MCDB	1.5	3.8	6.5	12.2	18.8	21.2	23.9	21.6	16.7	10.9	5.5	3.1		
	10%	WB	-0.2	0.4	2.4	6.2	11.5	14.5	17.5	16.7	13.0	8.1	3.5	0.9		
		MCDB	0.5	1.5	4.8	10.2	16.5	19.6	22.2	20.1	15.4	9.5	4.1	1.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.8	10.4	11.4	12.4	11.9	11.7	10.7	10.3	7.8	5.7	6.8		
		MCDBR	7.7	9.2	11.8	16.5	18.1	16.9	15.9	14.2	12.8	10.6	7.3	8.2		
		MCWBR	6.8	7.0	8.2	10.1	9.9	8.7	8.0	7.8	8.3	8.0	6.5	7.1		
		MCDBR	7.1	8.9	11.2	15.1	16.2	14.8	14.0	12.5	11.7	9.9	7.0	8.0		
	5% WB	MCWBR	6.5	7.0	7.9	9.6	9.4	8.0	7.5	7.3	7.9	7.9	6.4	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.231	0.288	0.314	0.351	0.355	0.349	0.365	0.357	0.323	0.308	0.248	0.347		
	taud		2.092	2.233	2.313	2.371	2.425	2.469	2.442	2.460	2.542	2.454	2.178	2.648		
	Ebn at Noon		439	664	793	836	864	875	848	823	792	652	415	273		
	Edh at Noon		32	62	83	97	100	98	98	88	68	51	29	20		
(44) All-Sky Solar Radiation	RadAvg		0.20	0.84	2.31	3.78	4.76	5.49	5.01	3.81	2.39	1.05	0.30	0.11		
	RadStd		0.03	0.13	0.25	0.32	0.49	0.36	0.56	0.32	0.29	0.18	0.06	0.01		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page